

Healthcare Data Science Report

Assignment 2 – Solution Proposal

Random forest method for predicting stroke probability ---- Based on age and other factors interacting

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Method Justification

Since the dataset used in this study is cross-sectional and population-based, it is impossible to accurately predict whether individual patients will have a stroke in the future. Therefore, constructing a machine learning model for predicting the risk of stroke at the individual level is neither appropriate nor may lead to significant errors. This project will attempt to build a machine learning model capable of identifying the risk characteristics of stroke at the population level, focusing on analyzing the differences in the incidence of stroke among different age groups (such as 0-30 years old, 31-50 years old, 51-70 years old, and 71 years old and above), and exploring whether there is a significant "age threshold" beyond which the risk of stroke increases significantly. This study will provide data support for the early screening of stroke and public health intervention.

The number of age variables for each research object in this data is relatively large, and there may be interactions among different variables. Machine learning is very effective in dealing with the extensive data in this regard [1]. Therefore, machine learning will be used as the research tool to construct a stroke prediction model mainly based on age analysis. Based on this, relevant literature retrieval was carried out according to the following PICO problems.

P (Patient / Population) : General adult population

I (Intervention / Exposure) : Machine learning models using age as a predictive feature

C (Comparison) : Conventional prediction methods (e.g., logistic regression or clinical risk scores)

O (Outcome) : Sensitivity and accuracy in predicting stroke occurrence

Previously, studies have used different machine learning models to predict strokes, namely seven different methods such as Gaussian Naive Bayes and logistic regression. Among them, AdaBoost, XGBoost and Random Forest classifiers have the lowest false prediction values, the highest accuracy rates and are the most reliable [2]. Gupta and Raheja (2022) pointed out the same view [3]. Reference [4] introduced some processing methods for missing values, solved the problems of missing data and category imbalance, and highlighted the effectiveness of ensemble learning technology and interpolation technology.

As analyzed in Literature [2], the study uses various machine learning algorithms (namely, Gaussian Naive Bayes, logistic regression, decision tree classifier, K-nearest neighbor, AdaBoost classifier, XGBoost classifier and random Forest classifier) to predict strokes and determine the most suitable model for predicting strokes. The data source is the stroke prediction dataset of Kaggle. The method is roughly divided into six parts, namely data preprocessing, data analysis, oversampling, training and test segmentation, model training and model evaluation. The data is divided in a ratio of 80:20, that is, the training set accounts for 80% of the total sample and is used for training the model. The test set accounts for 20% of the total sample and is used to test the model and conduct evaluations. It is concluded that the AdaBoost classifier, XGBoost classifier and Random forest classifier are all applicable to stroke prediction because they have the highest accuracy score, precision, recall rate, F1 score, AUC value and the fewest number of false predictions.

As analyzed in Literature [3], the research team proposed an integration technology based on the algorithm evaluation. The integration technology compiles predictions by using the research results of multiple basic models, and ultimately only yields one final result. Facts have proved that the integration method can better resist interference and overfitting, and by integrating the knowledge of multiple integration models, its effect is often better than that of a single model.

The analysis of Literature [4] is based on the dataset provided by the Kaggle platform. The study adopted multiple methods to correct the missing values and outliers, and scored the performance of different methods. The results show that the random forest algorithm has the best comprehensive performance, with a classification accuracy rate of 97.19% and a ROC curve score of 0.994. This indicates that the model has outstanding performance in both classification accuracy and category discrimination ability. The XGBoost algorithm also demonstrated outstanding performance, with its various indicators closely following.

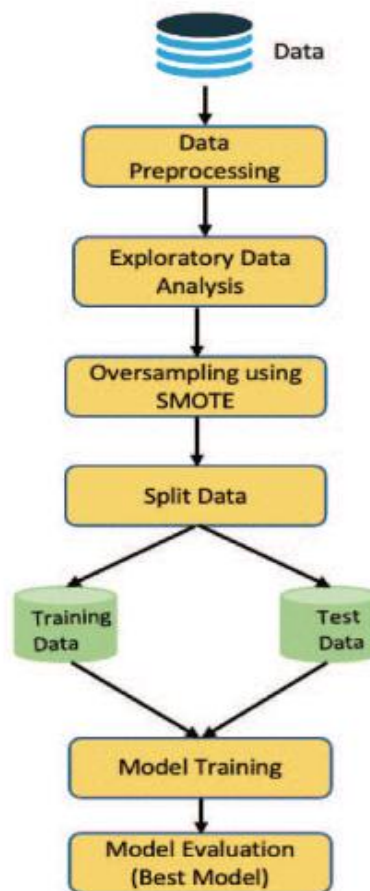
Given that this project aims to conduct predictive modeling using datasets containing multiple data types (such as continuous, discrete, ordinal and categorical), and most of these data have been converted into numerical forms through encoded dictionaries. Therefore, among numerous machine learning algorithms, the Random Forest model, as a reliable model with high accuracy, has the ability to handle a large number of features and shows better flexibility and accuracy than a single decision tree in classification tasks, especially when dealing with new data, it performs more stably and reliably. To sum up, combining the research results of the existing literature, the characteristics of the Kaggle dataset and the prediction goals that this project focuses on, the random forest model is finally selected as the main exploration tool of this study.

Implementation

According to the description in the previous section of this report, the dataset used in this project is derived from the Kaggle platform [5]. The data of this dataset has been de-identified and is publicly provided in CSV format, which can be directly imported into R for analysis. Therefore, this study does not require the approval of the Human Research Ethics Committee (HREC) when conducting exploratory data analysis (EDA) and developing machine learning models.

Furthermore, throughout the entire model development and future deployment process, the risk management framework for artificial intelligence in the United States [6] will always be referred to, similar to the ethical framework for artificial intelligence in Australia [7]. This move aims to ensure that the development and application process of the model complies with ethical norms, enhance public trust, and promote the responsible use of artificial intelligence technology.

Figure 1: Flow layout for Stroke Prediction[3]



Step 1: Data processing

For the Kaggle dataset to be analyzed, the uploader declares that data cleaning has been carried out and it is a clean dataset, which can skip the handling of missing values and outlier detection and processing. It should be noted that manual data type conversion and feature scaling operations may be required.

Step 2: EDA

Considering the definitions of the target/source/study population, the target population of this study is people suffering from stroke. Among them, their age will be the focus of the study, followed by the influence of the interaction between age and other variables on the probability of stroke.

In this dataset, descriptive statistical analysis and data visualization will be used to understand any pattern, examine the distribution and dispersion of variables, and explore the relationships between variables.

Especially in the process of exploratory data analysis (EDA), correlations in the data will be sought, and these correlations can be represented by scatter plots.

Step 3: Oversampling using SMOTH and Split Data

The Synthetic Minority Over-Sampling Technique is a commonly used oversampling technique, mainly used to deal with class imbalance problems. Unlike simply replicating minority class samples, the synthetic samples generated by SMOTE can increase data diversity, thereby reducing the risk of overfitting. The model can better learn the characteristics of minority class samples and thereby improve the accuracy of prediction. The data set is divided into two parts of 80% and 20% respectively, namely the training set and the test set, which can be used for cross-validation in the later stage.

Step 4: Model Training

The research will focus on using random forests as our classifiers and fitting them to the training data. According to the conclusion of the searched literature, random forest, as a training method of machine learning, is the most effective in predicting the risk of stroke.

Evaluation

Model evaluation is of vital importance, especially when dealing with such imbalanced datasets. In the analytical research, we employed more common indicators in the confusion matrix, such as precision, recall rate, F1 score and accuracy rate. This study mimics the evaluation process in Literature [4] and proposes some indicators for evaluating the model results to measure the reliability of the prediction results.

Accuracy: the proportion of all predictions that are correctly identified.

$$Accuracy = \frac{(\text{True positive} + \text{True negative})}{(\text{Total samples})}$$

Sensitivity: the proportion of all true positives, otherwise known as recall or the true positive rate (TPR).

$$Recall = \frac{\text{True positive}}{(\text{True positive} + \text{False negative})}$$

Specificity: the capacity of a system to produce accurate negative predictions, otherwise known as the true negative rate (TNR).

$$Specificity = \frac{\text{True negative}}{(\text{True negative} + \text{False positive})}$$

Precision: the ratio of correctly diagnosed positive samples among all positive samples.

$$\text{Precision} = \frac{\text{True positive}}{(\text{True positive} + \text{False positive})}$$

F-Measure: the harmonic mean of both precision and recall.

$$F - \text{Measure} = 2 \times \frac{\text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}}$$

Stakeholder's Interests

Prevention and Health Education:

At this stage, stakeholders can utilize patient data to create a heat map of stroke risk in the community and implement targeted interventions. Policy evaluation can assess the impact of policies on the incidence of stroke in specific populations by integrating multi-dimensional data. After patients input their personal information into the system, the trained model can automatically assess the risk of stroke, reducing the work pressure on the staff.

Acute Phase Treatment:

In acute treatment, the team can establish a stroke recurrence risk scoring model to improve the medication strategy and utilize gender and age factors to optimize the pre-screening process in the emergency department, thereby enhancing the treatment efficiency. The model can help doctors make etiological judgments more quickly and save precious rescue time.

Rehabilitation and Digital Long-Term Care:

The trained model can depict the profile of people prone to stroke, which is used to assist relevant practitioners in making judgments. The model helps hospitals identify high-risk patients, conduct key monitoring and intervention in advance, thereby reducing unnecessary waste of resources. By predicting the recovery progress and risks of patients, hospitals can adjust their rehabilitation plans based on the prediction results, identify patients who may need to extend their hospital stays in advance, reduce unnecessary hospital stays, and thereby save the operating costs of the hospital. Meanwhile, the prediction model provides a large amount of data support that enterprises can rely on. By analyzing patients' personal data, enterprises can develop personalized treatment plans for different patient groups and achieve more refined medical services.

Pharmaceutical Research and Innovation:

By leveraging machine learning, research institutions can accelerate the discovery and development process of drugs, thereby reducing the cost of drug research and development. Universities and enterprises can also enhance communication and cooperation through predictive models to create more job opportunities.

References

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